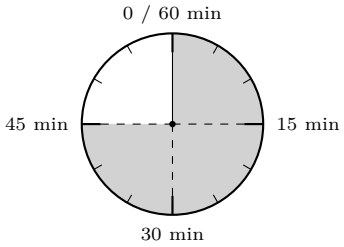
# Lesson 1: Time Math: Hours, Minutes and Timesheets

*Writing minutes as part of an hour, adding times, reading both kinds of clocks, and adding up the hours on a timesheet.*

## 1. Parts of an Hour: Fractions and Decimals

Pay is worked out in **hours**, but clocks show hours *and minutes*. So you need to write minutes as part of an hour. There are 60 minutes in 1 hour, so 15 min is $\frac{15}{60} = \frac{1}{4}$ of an hour.




The shaded part is 45 min. It is 3 of the 4 quarters of the clock, so

$$45 \text{ min} = \frac{45}{60} \text{ h} = \frac{3}{4} \text{ h} = 0.75 \text{ h}.$$

The same amount of time can be written three ways:

7 h 45 min = $7\frac{3}{4}$ h = 7.75 h

**Minutes to decimal hours:** divide the minutes by 60, then add the whole hours. 7 h 45 min: $45 \div 60 = 0.75$, so 7.75 h

**Decimal hours to minutes:** multiply only the decimal part by 60. 3.4 h: $0.4 \times 60 = 24$, so 3 h 24 min

**Benchmarks:** 15 min = $\frac{1}{4}$ h = 0.25 h 30 min = $\frac{1}{2}$ h = 0.5 h 45 min = $\frac{3}{4}$ h = 0.75 h

**Watch out.** The digits after a decimal point are *not* minutes. 7.30 h is not 7 h 30 min. 7 h 30 min = 7.5 h, and 7.30 h = 7 h 18 min (because $0.3 \times 60 = 18$).

**Calculator.** For 5 h 50 min, key `50` ÷ `60` + `5` ENTER. The screen shows `5.833333333`. The calculator divides before it adds, so you don't need brackets.

### Example 1

Complete the table of benchmarks. Write each fraction in simplest form. Round decimals to 2 places.

| **Minutes** | 15 | 30 | 45 | 20 | 40 | 10 | 6 | 12 |
|---|---|---|---|---|---|---|---|---|
| **Fraction of an hour** | $\frac{1}{4}$ | | | | | | | |
| **Decimal hours** | 0.25 | | | | | | | |

### Example 2

(a) Change to decimal hours. Round to 2 decimal places if needed.

(i) 3 h 15 min (ii) 5 h 50 min (iii) 8 h 36 min (iv) 95 min

(b) Change to hours and minutes.

(i) 6.5 h (ii) 4.75 h (iii) 2.4 h (iv) 1.35 h

Time is sometimes written as a fraction of an hour. A **mixed number**, like $2\frac{3}{4}$, has a whole number and a fraction. An **improper fraction**, like $\frac{11}{4}$, has a top (numerator) that is bigger than or equal to its bottom (denominator).

**Improper fraction to mixed number:** divide the top by the bottom. The answer is the whole number; the remainder goes over the bottom.
$\frac{9}{4}$: $9 \div 4 = 2$ remainder 1, so $\frac{9}{4} = 2\frac{1}{4}$ Then simplify the fraction if you can.

**Mixed number to improper fraction:** multiply the whole number by the bottom, add the top, keep the bottom.
$1\frac{1}{2} = \frac{1\times2+1}{2} = \frac{3}{2}$ (an hour and a half is 3 half-hours)

**Example 3**

(a) Write each time as a mixed number of hours.
(i) 2 h 45 min (ii) 1 h 20 min

(b) A mechanic's job card says a repair took $\frac{11}{4}$ h. Write this as a mixed number, and in hours and minutes.

(c) Ms. Okafor books swimming lessons that are each half an hour long. How many lessons fit in $3\frac{1}{2}$ h?

(d) A parent volunteer helped at the school for a total of 250 min. Write this in hours and minutes, as a mixed number, and in decimal hours.

### 2. Adding and Subtracting Hours and Minutes

Add hours to hours and minutes to minutes. An hour has 60 minutes, not 100, so you regroup at 60.

**Adding:** add the hours, add the minutes. If there are 60 min or more, trade 60 min for 1 h.
7 h 45 min + 6 h 30 min = 13 h 75 min = 14 h 15 min (75 min = 1 h 15 min)

**Subtracting:** if there are not enough minutes on top, borrow 1 h and add 60 min.
5 h 10 min − 2 h 40 min $\rightarrow$ 4 h 70 min − 2 h 40 min = 2 h 30 min

**Watch out.** Don't type 7.45 + 6.30 into your calculator. 7 h 45 min is not 7.45 h. Either add the hours and minutes by hand, or change each time to decimal hours first ($7.75 + 6.5 = 14.25$ h).

**Example 4**

(a) 4 h 25 min + 3 h 50 min

(b) 2 h 40 min + 1 h 35 min + 3 h 55 min

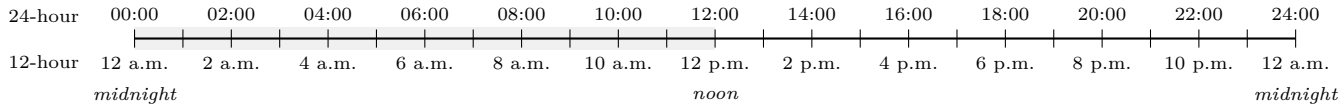
(c) 9 h 15 min − 3 h 40 min

(d) A delivery company lets its drivers drive at most 11 h in a day. Rosa has driven 4 h 35 min this morning and 3 h 50 min this afternoon. How much longer can she drive today?

### 3. The 12-Hour and 24-Hour Clock

The **12-hour clock** counts from 12 to 12 twice a day. **a.m.** times run from midnight to noon. **p.m.** times run from noon to midnight. 12:00 a.m. is midnight and 12:00 p.m. is noon.

The **24-hour clock** counts the hours from 00 to 23 and has no a.m. or p.m. It always uses two digits for the hour: 07:30 is 7:30 a.m. and 19:30 is 7:30 p.m. Hospitals, airlines, the military, transit schedules and many work schedules use it.




**12-hour to 24-hour:** a.m.: keep the hour (write 2 digits): 9:15 a.m. = 09:15.
p.m.: add 12 to the hour: 4:20 p.m. = 16:20.
The two 12 o'clocks are different: 12:40 a.m. = 00:40 (just after midnight) 12:40 p.m. = 12:40 (just after noon)

**24-hour to 12-hour:** hours 13 to 23: subtract 12, p.m. hour 12: p.m. hour 00: write 12, a.m. hours 01 to 11: a.m.

**Example 5**

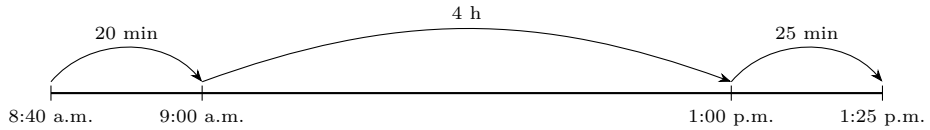
Complete the table.

| **12-hour clock** | 6:15 a.m. | 3:40 p.m. | 12:20 a.m. | 12:05 p.m. | 11:55 p.m. |
|---|---|---|---|---|---|
| **24-hour clock** | | | | | |
| **24-hour clock** | 08:30 | 17:10 | 00:45 | 21:00 | 12:30 |
| **12-hour clock** | | | | | |

### 4. Elapsed Time

**Elapsed time** is how much time passes from a start time to an end time. You can't just subtract the clock times on a calculator, because the minutes go up to 60, not 100. Instead, **count up** from the start time in friendly jumps.

**Count up:** Step 1: jump to the next whole hour. Step 2: jump the whole hours. Step 3: jump the minutes left. Step 4: add the jumps.




From 8:40 a.m. to 1:25 p.m.: 20 min + 4 h + 25 min = **4 h 45 min**.

**Or use the 24-hour clock and subtract:** 13:25 − 08:40 → borrow 1 h: 12:85 − 08:40 = 4 h 45 min.

**Across midnight:** find the time up to midnight, then add the time after midnight.

**Watch out.** The time from 8:50 to 10:15 is *not* 10.15 − 8.50 = 1.65. There are 60 minutes in an hour, so count up: 10 min + 1 h + 15 min = 1 h 25 min.

**Example 6**

Find the elapsed time.

(a) from 8:50 a.m. to 11:20 a.m.

(b) from 10:35 a.m. to 3:10 p.m. (Do it two ways.)

**Example 7**

(a) A security guard works overnight from 10:45 p.m. to 7:15 a.m. How long is his shift? Give the answer in hours and minutes and in decimal hours.

(b) Farah takes the bus from Calgary to Edmonton. It leaves at 13:50 and the trip takes 3 h 35 min. When does it arrive? Give the time in both clocks.

### 5. Timesheets and Unpaid Breaks

A **timesheet** (or **time card**) records when you start work (**in**) and when you stop (**out**) each day. Your employer uses it to work out your pay, so it must be right. You are not paid for an **unpaid break**, such as a 30 min lunch, so subtract it. Some jobs have a **split shift**: two blocks of work in one day, with time off in between. Find the time for each block, then add.

**Hours worked in a day** = (time from in to out) − unpaid break

**Hours for the week:** add the days in hours and minutes. Change the total to decimal hours *at the end.*

**Did you know?.** In Alberta, a worker must get a break of at least 30 min during a shift of more than 5 hours in a row. The employer decides whether the break is paid. (Rules as of 2026; rules can change.)

**Example 8**

(a) Priya works at a grocery store from 8:30 a.m. to 5:15 p.m. She takes a 30 min unpaid lunch. How many hours does she work? Give the answer in hours and minutes and in decimal hours.

(b) Mr. Dhillon drives a school bus. He works a split shift: 6:45 a.m. to 9:10 a.m., and 2:50 p.m. to 5:05 p.m. How many hours does he work? Round decimal hours to 2 places.

**Example 9**

Jordan is a cook at a restaurant in Cochrane. Complete his timesheet for the week.

| **Day** | **In** | **Out** | **Time in to out** | **Unpaid break** | **Hours worked** |
|---|---|---|---|---|---|
| Monday | 7:15 a.m. | 3:45 p.m. | | 30 min | |
| Tuesday | 11:30 a.m. | 7:50 p.m. | | 30 min | |
| Wednesday | day off | | | | |
| Thursday | 4:00 p.m. | 11:45 p.m. | | 45 min | |
| Friday | 5:30 p.m. | 1:15 a.m. | | 30 min | |
| Saturday | 9:00 a.m. | 1:40 p.m. | | none | |
| | | | | **Total hours** (hours and minutes) | |
| | | | | **Total hours** (decimal hours) | |

## Lesson 2: Wages and Salaries: Gross Pay and Pay Periods

*How much you earn before anything is taken off, whether you are paid by the hour or by the year.*

### 1. Wages, Salaries and Gross Pay

Your **income** is the money you are paid for the work you do. Two of the most common ways to be paid are:

- a **wage**: you are paid a set amount for each hour you work (an **hourly wage** or **hourly rate**, like \$17.50/h). More hours means more pay.
- a **salary**: you are paid a set amount for a whole year (like \$52 000 a year), split into equal paycheques. The pay stays the same even if you work a little more or less one week.

**Gross pay** is everything you earn *before* anything is taken off. Your employer takes money off your cheque for things like income tax. These are called **deductions**. The amount you actually take home is your **net pay**, so net pay is always less than gross pay. (Deductions are in Lesson 7.)

| Jobs that often pay a wage | Jobs that often pay a salary |
|---|---|
| cashier, cook, server, warehouse worker, lifeguard, construction labourer, apprentice in a trade | teacher, office manager, accountant, engineer, store manager, police officer |

### 2. Gross Pay from an Hourly Wage

To find gross pay for an hourly job, multiply the hours by the hourly wage.

**Gross pay from a wage** gross pay = hours worked × hourly wage

Write the hours as a **decimal** first. Round the pay to the nearest cent at the end.

**Quick refresher.** minutes ÷ 60 = part of an hour. 15 min = 0.25 h, 30 min = 0.5 h, 45 min = 0.75 h. So 7 h 45 min = 7.75 h, not 7.45 h.

**Estimate.** Round the hours and the wage to easy numbers. For 7.75 h at \$23.85/h: about 8 × \$24 = \$192. If your calculator says \$1848 or \$18.48, you know something went wrong.

**Example 1**

(a) Noor works at a garden centre for \$16.40/h. On Saturday she works 6 hours. What is her gross pay?

(b) Dev is an apprentice carpenter. He earns \$23.85/h. One day he works 7 h 45 min. Estimate, then find his gross pay.

Most hourly workers fill in a **timesheet** (or time card). It shows when each shift starts and ends. A break you are *not* paid for (an **unpaid break**) is taken off the hours.

**Example 2**

Sofia is a cook at a café in Cochrane. She earns \$17.60/h. Complete her timesheet, then find her gross pay for the week.

| Day | Start | End | Unpaid break | Hours paid |
|---|---|---|---|---|
| Monday | 6:00 a.m. | 2:30 p.m. | 30 min | |
| Tuesday | 7:00 a.m. | 12:00 p.m. | none | |
| Wednesday | day off | | | 0 |
| Thursday | 10:30 a.m. | 4:45 p.m. | 30 min | |
| Friday | 11:00 a.m. | 7:30 p.m. | 30 min | |
| Saturday | 8:00 a.m. | 12:00 p.m. | none | |
| | | | **Total hours** | |

If you know the pay and the hours, you can find the hourly wage. Undo the multiplying: divide.

**Finding the hourly wage** hourly wage = gross pay ÷ hours worked

**Example 3**

Amara works at a pharmacy. Last week she worked 25.5 hours and her gross pay was $446.25. What is her hourly wage? Check your answer.

### 3. Alberta's Minimum Wage

The **minimum wage** is the lowest hourly wage an employer is allowed to pay. Each province sets its own.

**Alberta minimum wage** (as of 2026; rates can change)

- Most workers: at least **$15.00/h**.
- Students **under 18**: can be paid **$13.00/h** for the **first 28 hours** in a week *while school is in session*. Any hours past 28 that week, and all hours in the summer or other school breaks, must be paid at least $15.00/h.

To check a job, find the hourly wage (pay ÷ hours) and compare it with the minimum.

**Example 4**

Is each person paid at least Alberta's minimum wage?

(a) Ben is 22. He earned $456.00 for 32 hours at a car dealership.

(b) Lucy is 16. In November she earned $283.50 for 21 hours at a bakery.

(c) In July, Lucy earned $378.00 for 28 hours at the same bakery.

**Example 5**

Tyler is 17 and works at a sporting goods store. His employer pays the lowest wage the law allows. In one week in October, Tyler works 31 hours. What is his gross pay?

### 4. Salaries and Pay Periods

A salary is a yearly amount, but nobody waits a whole year to be paid. The salary is split into equal paycheques. The time each cheque covers is a **pay period**.

| Pay period | How often you are paid | Paycheques in a year |
|---|---|---|
| weekly | every week | 52 |
| biweekly | every two weeks | 26 |
| semi-monthly | twice a month (e.g. the 15th and the last day) | 24 |
| monthly | once a month | 12 |

| | |
|---|---|
| **Salaries** | pay per cheque = yearly salary ÷ number of pay periods<br>yearly salary = pay per cheque × number of pay periods |

**Watch out.** *Bi*weekly (every 2 weeks, 26 cheques) and *semi*-monthly (twice a month, 24 cheques) sound alike but are different. A month is a bit more than 4 weeks, so 26 is more than $2 \times 12 = 24$.

**Example 6**

Hana is a dental office manager. Her salary is \$54 600 a year.

(a) Complete the table to show her gross pay for each pay period.

| Pay period | weekly | biweekly | semi-monthly | monthly |
|---|---|---|---|---|
| Calculation | | | | |
| Gross pay | | | | |

(b) Marco is a lab technician. He is paid \$1925.50 biweekly. What is his yearly salary? What is his average pay per month?

To compare a salary with an hourly wage, find the salary's **hourly equivalent**: what it works out to for each hour worked. A full-time 40-hour week is $40 \times 52 = 2080$ hours a year.

| | |
|---|---|
| **Hourly equivalent of a salary** | hourly equivalent = yearly salary ÷ (hours per week × 52) |
| **Yearly pay from a wage** | yearly pay = hourly wage × hours per week × weeks worked |

**Calculator.** For \$46 800 and 40 h a week: 46800 [÷] [(] 40 [×] 52 [)] [ENTER] gives 22.5. Without the brackets the calculator divides by 40 and then *multiplies* by 52.

**Example 7**

(a) A youth centre pays its program coordinator a salary of \$46 800 for a 40-hour week. What is the hourly equivalent?

(b) Jesse works at a print shop for \$21.40/h, 37.5 hours every week of the year. What is his yearly gross pay?

### 5. Comparing Wage and Salary Jobs

To compare two jobs fairly, put them in the **same units**: pay per week (or per year) *and* pay per hour. Then think about more than the money.

| | **Hourly wage** | **Salary** |
|---|---|---|
| **Good** | paid for every hour you work; extra hours mean extra pay (and overtime) | the same pay every cheque, so it is easy to budget; often paid vacation and benefits |
| **Not so good** | hours can be cut, so pay can change; no work means no pay | extra hours usually mean no extra pay |

**Example 8**

Riya has two job offers.

- **Job A:** \$22.75/h, 35 hours a week.
- **Job B:** a salary of \$42 900 a year, 37.5 hours a week.

(a) Find the weekly pay for each job.

(b) Find the pay per hour for each job.

(c) What if Job A gave Riya 40 hours a week instead? Which job would pay more each week then?

(d) Which job would you pick? Give one reason about money and one other reason.

**Example 9**

Find the mistake in each solution and fix it.

(a) Cole's salary is \$39 000 a year, paid semi-monthly.
*Cole's work:* \$39 000 ÷ 26 = \$1500.00 per cheque.

(b) Erin worked 6 h 45 min at \$16.00/h.
*Erin's work:* 6.45 × \$16.00 = \$103.20.

## Lesson 3: Overtime: Time-and-a-Half and Double Time

*What you earn for extra hours, and how Alberta decides which hours count as overtime.*

### 1. Overtime Rates

Every job has a usual number of hours, called the **regular hours**. Hours worked past that are **overtime**. Overtime hours are paid at a higher rate than your regular hourly wage (your **base wage**).

- **Time and a half**: 1.5 times your base wage. This is the usual overtime rate.
- **Double time**: 2 times your base wage. Some employers and union agreements pay this, for example on holidays or after a very long day. Use it only when a question says the job pays it.

A few agreements use other numbers, such as 1.25 times or 2.5 times ("double time and a half"). They work the same way: multiply the base wage by the number.

**Overtime rates**

time-and-a-half rate = base wage × 1.5 double-time rate = base wage × 2

overtime pay = overtime hours × overtime rate

**Watch out.** Time and a half means 1.5 *times* the wage. It does not mean adding \$1.50 to the wage, or adding half an hour. At \$18.00/h, time and a half is \$18.00 × 1.5 = \$27.00/h, not \$19.50/h.

**Estimate.** Half of the wage, added to the wage, gives time and a half. Half of \$18 is \$9, and \$18 + \$9 = \$27.

**Example 1**

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Complete the table. Alberta's minimum wage is \$15.00/h (as of 2026; rates can change).

| **Base wage** | \$15.00/h | \$18.50/h | \$21.20/h | \$27.80/h |
|---|---|---|---|---|
| **Time and a half** (× 1.5) | | | | |
| **Double time** (× 2) | | | | |

### 2. Gross Pay With Overtime

When you work overtime, your **gross pay** (pay before deductions) has two parts: **regular pay** for the regular hours and **overtime pay** for the extra hours.

**Gross pay with overtime**

Step 1: regular pay = regular hours × base wage

Step 2: overtime pay = overtime hours × overtime rate

Step 3: gross pay = regular pay + overtime pay

Round to the nearest cent at the *end*.

The job sets when overtime starts. Many employers pay overtime after 40 hours a week. The law in Alberta sets the *latest* point it can start (Section 3). An employer may always pay overtime sooner, because that is better for the worker. If a question gives the employer's rule, use it.

**Example 2**

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Jordan is a line cook. He earns \$17.60/h. Last week he worked 40 regular hours and 6 overtime hours at time and a half. Find his gross pay.

**Calculator.** If the overtime rate has three decimals, do not round it. Do the overtime pay in one line instead: 6.5 [×] 1.5 [×] 22.75 [ENTER] shows 221.8125. Round only the final answer.

**Example 3**

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Priya is a drywall labourer. She earns \$22.75/h. Her company pays time and a half for every hour over 40 in a week. This week she worked 46 h 30 min. Find her gross pay.

**Example 4**

Some agreements pay double time.

(a) Kyle works at a grocery store. His union agreement pays double time on general holidays. He earns $16.20/h and works a 7.5 h shift on Canada Day. How much does he earn for the shift?

(b) Ayesha is an apprentice electrician. She earns $28.60/h. One week she worked 44 regular hours, 4 hours at time and a half and 3 hours at double time. Find her gross pay.

### 3. Alberta's Overtime Rule: 8 Hours a Day or 44 Hours a Week

In Alberta, the law says most workers must get overtime at time and a half for:

- every hour over **8 hours in a day** (**daily overtime**), or
- every hour over **44 hours in a week** (**weekly overtime**),

**whichever gives more overtime hours.** You work out both, then use the bigger number. (Rules as of 2026. Rules can change, and a few kinds of jobs have different rules.)

**Alberta 8/44 rule**

Step 1: **Daily overtime.** For each day, hours over 8. Add them up.

Step 2: **Weekly overtime.** Total hours for the week − 44. (If the total is 44 or less, this is 0.)

Step 3: **Overtime hours** = the *bigger* of the two. regular hours = total hours − overtime hours

Step 4: Find the gross pay (regular pay + overtime pay).

**Watch out.** Do not add the daily and weekly overtime together. They count the same extra hours in two ways. Pick the bigger one.

**Example 5**

Maya works at a tire shop for $18.00/h. Complete her timesheet. Then find her overtime hours and her gross pay.

| **Day** | **Hours** | **Hours over 8** |
|---|---|---|
| Monday | 10 | |
| Tuesday | 8 | |
| Wednesday | 9.5 | |
| Thursday | 7 | |
| Friday | 8 | |
| **Total** | | |

**Example 6**

Ethan stocks shelves at a hardware store for $16.50/h. He worked Monday to Saturday: 8 h, 8 h, 8.5 h, 8 h, 8 h and 7.5 h. Find his overtime hours and his gross pay.

**When does each rule win?.** Long days (more than 8 hours) make daily overtime bigger. Many days in a row, even short ones, can make weekly overtime bigger. That is why you always check both.

### 4. From Timesheet to Paycheque

Real timesheets show start and end times, not hours. Put together what you learned in Lesson 1 and in this lesson:

Step 1: For each day, find the time from start to end, and subtract any **unpaid break**.
Step 2: Write each day's hours as a decimal (15 min = 0.25 h, 30 min = 0.5 h, 45 min = 0.75 h).
Step 3: Use the 8/44 rule to find the overtime hours, then the gross pay.

**Example 7**

Tomás is a landscaper in Cochrane. He earns $19.20/h. Complete his timesheet. Use the Alberta 8/44 rule to find his gross pay.

| Day | Start | End | Unpaid break | Hours worked | Hours over 8 |
|---|---|---|---|---|---|
| Monday | 7:00 a.m. | 5:30 p.m. | 30 min | | |
| Tuesday | 7:00 a.m. | 3:30 p.m. | 30 min | | |
| Wednesday | 6:30 a.m. | 5:15 p.m. | 45 min | | |
| Thursday | 7:00 a.m. | 3:00 p.m. | 30 min | | |
| Friday | 7:00 a.m. | 4:15 p.m. | 30 min | | |
| | | | **Total** | | |

## Lesson 4: Piecework and Contracts

*Getting paid for each piece you finish, or one set price for a whole job, and working out what that pay is worth per hour.*

### 1. Piecework: Paid for Each Piece

People earn money in many ways. You already know two: an **hourly wage** (paid for each hour) and a **salary** (a set amount per year).

| Method | You are paid . . . | Jobs that often use it |
|---|---|---|
| Hourly wage | for each hour you work | cashier, cook, labourer, lifeguard |
| Salary | a set amount per year, split into pay periods | teacher, nurse, office manager |
| **Piecework** | for each piece or task you finish | tree planter, fruit picker, flyer carrier, sewing worker |
| **Contract** | one set price for a whole job | house painter, roofer, landscaper, web designer |
| Commission (Lesson 5) | a percent of what you sell | car salesperson, real estate agent |
| Tips and bonuses (Lesson 6) | extra money on top of your pay | server, hairstylist, hotel staff |

With **piecework**, you are paid a set amount, the **piece rate**, for each item you make or each task you finish. The more pieces you finish, the more you earn.

**Piecework pay** = piece rate × number of pieces  e.g. 2350 seedlings at \$0.16 each: $2350 \times \$0.16 = \$376.00$

**Calculator.** 2350 [×] 0.16 [ENTER] shows 376. Write it as money: \$376.00. Estimate to check: about 2000 trees at about 20¢ is about \$400.

**Example 1**

Find each worker's pay for the day. The tree planter is Nadia.

| Worker | Pieces | Piece rate | Work (rate × pieces) | Pay |
|---|---|---|---|---|
| (a) Tree planter | 2350 seedlings | \$0.16 each | | |
| (b) Flyer carrier | 580 flyers | \$0.05 each | | |
| (c) Sewing worker | 64 pillow covers | \$1.85 each | | |
| (d) Strawberry picker | 27 flats | \$4.25 per flat | | |

**Different rates for different pieces.** Harder pieces often pay more. Find the pay for each kind of piece, then add.

**Example 2**

Mei-Ling does clothing alterations at a dry cleaner in Airdrie. She is paid by the piece.

| Task | Rate | Number | Pay |
|---|---|---|---|
| Hem a pair of pants | \$9.50 | 18 | |
| Replace a zipper | \$14.00 | 7 | |
| Take in a shirt | \$11.25 | 4 | |
| | | **Total for the week** | |

(a) Complete the table.

(b) Mei-Ling did all this work in 13.5 hours. What did she earn per hour? Round to the nearest cent.

**What is piecework worth per hour?** To compare piecework with an hourly job, divide the pay by the hours worked. This is the **effective hourly rate**: what the pay works out to for each hour (like the hourly equivalent of a salary in Lesson 2).

**Effective hourly rate** = pay ÷ hours worked

**Minimum wage.** In Alberta, most employees who are paid by the piece must still earn at least the **minimum wage**, \$15.00 per hour, for the hours they work (as of 2026; rates can change). If the piecework pay is less, the employer must add the difference.

**Example 3**

Nadia is the tree planter from Example 1. She worked 10 hours to plant the 2350 seedlings.

(a) What was her effective hourly rate?

(b) On a rainy day, she plants only 820 seedlings in 9 hours. Find her effective hourly rate. Does her employer need to add money to reach minimum wage? If so, how much?

### 2. Working Backwards with Piecework

Sometimes you know the pay and want the piece rate, or you know the pay you want and need the number of pieces. The formula can be turned around.

**pay** = rate × pieces **rate** = pay ÷ pieces **pieces** = pay ÷ rate

Check by multiplying: rate × pieces should give the pay.

**Watch out.** When you find how many pieces you need to reach a goal, **round up**. You cannot get paid for part of a piece, and rounding down leaves you short of your goal.

**Example 4**

(a) A florist is paid \$382.50 to make 30 table centrepieces for a wedding. How much is she paid for each one?

(b) Tomas translates documents for \$0.12 per word. He was paid \$258.00 for one document. How many words long was it?

(c) Sana picks strawberries for \$4.25 per flat. How many flats must she pick to earn at least \$120.00?

**The employer's side.** An employer who pays by the piece then sells the pieces for more. The difference pays for the employer's other costs (land, tools, rent) and profit.

**Example 5**

A berry farm near Red Deer pays pickers \$3.50 for each basket of raspberries. The farm sells each basket at its market stand for \$9.00. One day the pickers fill 240 baskets, and the farm sells them all.

(a) How much does the farm pay the pickers?

(b) How much money does the farm take in, and how much is left after paying the pickers?

(c) What percent of the selling price goes to the picker? Round to one decimal place.

### 3. Contracts: One Price for the Whole Job

A **contract** is an agreement to do a whole job for one set price, no matter how long it takes. The person who does the job is a **contractor**. Before the job, a contractor gives the customer a **bid** (or **quote**): the price they will charge.

Out of the contract price, the contractor usually pays for the **materials** (paint, lumber, parts) and for any **helpers**. What is left is what the contractor really earns.

**Contractor's earnings** = contract price − materials − helpers' wages − other costs
**Effective hourly rate** = contractor's earnings ÷ hours the contractor worked
**Helpers' wages** = number of helpers × hours × hourly wage

**Watch out.** Subtract all the costs **before** you divide by the hours. The contract price is not all yours.

**Calculator.** Use brackets: [(] 2400 [−] 465.80 [)] [÷] 38 [ENTER]. Or find the earnings first, press [ENTER], then [÷] 38 [ENTER].

**Example 6**

Darnell is a house painter. He agrees to paint the inside of a house in Cochrane for $2400.00. He buys the paint and supplies for $465.80. He does all the work himself.

(a) The job takes Darnell 38 hours. How much does he earn, and what is his effective hourly rate?

(b) Suppose the walls need extra repairs, and the job takes 52 hours instead. The price stays the same. What is his effective hourly rate now?

**Setting a contract price.** To make a bid, the contractor adds up all the costs and the amount they want to earn. Some contractors instead add a percent to their costs for their profit, the same way a store adds a markup (Unit 1).

**Contract price** = materials + helpers' wages + what the contractor wants to earn

or **contract price** = costs × (1 + profit rate) e.g. costs plus 30%: costs × 1.30

**Example 7**

Amara runs a small landscaping business in Okotoks. She is bidding on a job to build a stone patio.

- Pavers, gravel and sand cost $1180.00.
- She will hire 2 helpers for 12 hours each at $19.50 per hour.
- She wants to earn $600.00 for herself.

(a) What should Amara charge for the patio?

(b) Amara's friend says, "Just charge your costs plus 30%." What price would that give? How much would Amara earn for herself?

### 4. Which Is Better? Comparing Ways to Earn

Each way of being paid has good points (**advantages**) and bad points (**disadvantages**) for the worker.

| Method | Advantages | Disadvantages |
|---|---|---|
| Hourly wage | Paid for every hour, even on slow days. Overtime pay for long days and weeks. | Working faster does not earn more money. |
| Piecework | Fast, skilled workers earn more. You can see your pay grow as you work. | Pay drops on slow days (bad weather, learning a new job). Rushing can lead to mistakes or injuries. |
| Contract | If you work well and fast, you can earn a lot per hour. You are your own boss and choose your jobs. | If the job takes longer or costs more, you earn less. You pay for materials, tools and helpers. No minimum wage, overtime, paid vacation or benefits. You must keep finding new jobs. |

To compare a contract with an hourly job, find what each pays for the **same number of hours**. Ask "what if the job takes longer?"

**Example 8**

Jaxon installs drywall. A builder offers him two ways to be paid for finishing a basement. The work is spread over two weeks, so there is no overtime.

- Option A: $26.50 per hour.
- Option B: a contract for $1400.00. Jaxon must buy the tape and mud, about $95.00.

(a) Jaxon thinks the job will take 40 hours. Which option pays more, and by how much?

(b) What if the job takes 55 hours? Which option pays more now?

(c) For about how many hours do the two options pay the same? Round to one decimal place.

**Example 9**

Kiana is choosing between two jobs at a sewing company.

- Job 1: $16.50 per hour.
- Job 2: piecework, $2.40 for each bag sewn (the company tops up to the $15.00 minimum wage if needed).

As a beginner, Kiana can sew about 5 bags an hour. After a few months of practice, she expects to sew about 9 bags an hour.

(a) Find what Job 2 pays per hour for a beginner and after practice.

(b) Finish the sentences.

As a beginner, the better job is _____ because _____.

After practice, the better job is _____ because _____.

One disadvantage of Job 2 is _____

## Lesson 5: Commission and Salary Plus Commission

*Jobs that pay you a share of what you sell: straight commission, salary plus commission, graduated rates, quotas, and how to compare pay plans.*

### 1. Straight Commission

Some jobs pay you for what you *sell*, not for your time. A **commission** is money you earn as a percent of your sales. The percent is the **commission rate**. Your **sales** are the total dollar value of everything you sold.

If commission is your *only* pay, it is called **straight commission**. Jobs that often pay this way: real estate agents, car and RV salespeople, and some furniture, appliance and travel sales jobs.

| | |
|---|---|
| **Commission** | commission = sales × commission rate (change the rate to a decimal first) |
| e.g. 5% commission on \$2000 of sales: | $\$2000 \times 0.05 = \$100$ |

**Watch out.** Write the rate as a decimal carefully: $6\% = 0.06$, $2.5\% = 0.025$, $12\% = 0.12$. A commission is a *part* of the sales, so it should be much smaller than the sales.

**Estimate.** Find 10% first (move the decimal one place left). 5% is half of that. For 6%, the answer is a bit more than the 5% amount.

**Example 1**

Maya sells appliances (fridges, stoves, washers) at a store in Calgary. She earns 6% commission. Last week she sold \$7385.00 worth of appliances.

(a) Estimate her commission.

(b) Calculate her commission.

**Calculator.** 7385 [×] 0.06 [ENTER] shows 443.1. Write it as money: \$443.10.

**Example 2**

(a) Kai sells trailers at an RV lot in Airdrie. He earns 2.5% of the selling price. He sells a travel trailer for \$38 900.00. What is his commission?

(b) Bianca works in a shoe store. She earns 8% commission. Her sales this week were \$1249.99, \$869.50 and \$2380.00. What is her commission for the week?

### 2. Salary Plus Commission

Straight commission is risky: in a slow week you might earn very little. Many sales jobs pay a **base pay** that you always get, *plus* a commission. The base pay can be a **salary** (a set amount each pay period) or an **hourly wage** (hours × rate, as in Lesson 2).

| | |
|---|---|
| **Salary plus commission** | gross pay = base pay + commission |
| | gross pay = base pay + (sales × commission rate) |
| Base pay is a salary, or hours worked × hourly wage. | |

**Watch out.** Find the commission first, then add the base pay. Do not take the percent of the base pay.

**Example 3**

(a) Jordan sells cell phone plans. He earns $450.00 a week plus 4% of his sales. This week he sold $7325.00. Find his gross pay.

(b) Priya works at a mattress store in Lethbridge. She earns $15.50/h plus 2.5% commission. Last week she worked 32 h and sold $18 640.00. Find her gross pay.

### 3. Graduated Commission and Quotas

Some employers pay a *higher* rate when you sell more. In a **graduated commission** (also called a **tiered** commission), the sales are split into levels, and each level has its own rate. For example: 3% on the first $10 000 of sales, and 5% on the amount *over* $10 000.

**Graduated commission**
Step 1: Split the sales into the levels. Step 2: Find the commission for each level. Step 3: Add.

**Watch out.** The higher rate is only for the part of the sales *above* the level, not for all the sales.

**Example 4**

Lucas sells furniture. He earns 3% on the first $10 000 of his monthly sales and 5% on sales over $10 000. In March he sold $16 400. Complete the table to find his commission.

| Level | Sales in this level | Rate | Commission |
|---|---|---|---|
| First $10 000 | | 3% | |
| Over $10 000 | | 5% | |
| | | **Total commission** | |

**Calculator.** Do it in one line with brackets: 10000 [×] 0.03 [+] [(] 16400 [−] 10000 [)] [×] 0.05 [ENTER] shows 620.

A **quota** is a sales target. Some jobs pay commission only on the sales *over* the quota. Sell less than the quota and you get only your base pay.

**Example 5**

Hana sells advertising for a radio station in Red Deer. She earns $2400.00 a month plus 6% of her sales over a monthly quota of $20 000.

(a) Find her gross pay in a month when she sells $27 850.

(b) Find her gross pay in a month when she sells $18 500.

### 4. Finding the Rate or the Sales

Sometimes you know the commission and want the rate or the sales. Use the same formula, worked backwards. (Lesson 8 does more of this.)

**Working backwards**

$$\text{commission rate} = \frac{\text{commission}}{\text{sales}} \times 100 \quad \text{(as a percent)} \qquad \text{sales} = \frac{\text{commission}}{\text{rate as a decimal}}$$

Check: sales × rate should give the commission.

**Example 6**

(a) Ethan earned \$378.00 in commission on sales of \$8400.00. What is his commission rate?

(b) A band's booking agent keeps 15% of what the band is paid for each show. Last summer the agent earned \$1620.00 from the band's shows. How much was the band paid in all?

### 5. Comparing Pay Plans

Which pays more: a steady salary or a plan with commission? It depends on how much you sell. Make a table of the pay at a few sales amounts, then look for where the plans pay the same.

**Where two plans pay the same** (salary only vs. smaller salary + commission)

sales needed = (difference in base pay) ÷ (commission rate as a decimal)

Sell more than this and the commission plan pays more. Sell less and the salary pays more.

**Example 7**

Ava has two job offers selling windows and doors for a home renovation company.

**Job A:** \$2800 a month salary. **Job B:** \$1500 a month plus 5% of her sales.

(a) Complete the table.

| **Monthly sales** | \$10 000 | \$20 000 | \$30 000 | \$40 000 |
|---|---|---|---|---|
| **Job A pay** | | | | |
| **Job B pay** | | | | |

(b) At what monthly sales do the two jobs pay the same?

(c) Ava thinks she can sell about \$24 000 a month. Which job should she take? Give a reason that is not only the pay.

**Advantages and disadvantages.** Complete the table with the class.

| Pay plan | An advantage for the worker | A disadvantage for the worker | Jobs that use it |
|---|---|---|---|
| Salary or wage only | | | |
| Straight commission | | | |
| Salary + commission | | | |

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## Lesson 6: Extra Earnings: Tips, Bonuses and Shift Premiums

*Money you can earn on top of your wage or salary, and how to add it all up to find your gross pay.*

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### 1. Tips

Many jobs pay extra money on top of the base wage or salary. All of this extra money is part of your **gross pay** (your pay before deductions).

A **tip** (also called a **gratuity**) is money a customer gives to thank a worker for good service. Servers, bartenders, hairstylists, taxi and delivery drivers often earn tips. A tip is usually a percent of the bill. In Canada, 15% to 20% is common. In this booklet, the tip is found on the bill *before* tax.

In Alberta, a server must still be paid at least minimum wage ($15.00/h as of 2026; rates can change). The tips are extra, and they belong to the workers, not to the owner.

**Tips**

$\text{tip} = \text{bill} \times \text{tip rate}$ $\qquad$ $\text{tip percent} = \dfrac{\text{tip}}{\text{bill}} \times 100$

$\text{gross pay} = \text{wage pay} + \text{tips}$ $\qquad$ $(\text{wage pay} = \text{hours} \times \text{hourly wage})$

**Estimate.** Round the bill. Find 10% by moving the decimal one place left. Then 15% = 10% + half of 10%, and 20% = 2 × 10%.

**Example 1**

(a) A family's dinner bill in Calgary is $64.80 before tax. Estimate a 15% tip in your head. Then find a 15% tip and a 20% tip.

(b) A customer leaves a $9.00 tip on a $52.00 haircut. What percent tip is that? Round to the nearest tenth of a percent.

**Example 2**

Noor is a server at a restaurant in Lethbridge. She earns \$15.50/h. On Friday she worked a 6-hour shift. Her customers' bills added up to \$1240.00, and her tips were 16% of the bills.

(a) Find Noor's gross pay for the shift.

(b) How much did she earn per hour, with tips?

Sometimes you know the tips and the tip rate, and you want the total of the bills. Work backwards: divide the tips by the rate.

| **Working backwards** | bill = tip ÷ tip rate | e.g. \$30 is 15% of the bill: \$30 ÷ 0.15 = \$200 |
|---|---|---|

**Example 3**

Mateo delivers food in Airdrie. Last weekend he got \$86.40 in tips. His tips were 12% of the orders he delivered, on average. What was the total of his orders?

**Watch out.** Tips change from day to day. A busy Saturday and a slow Tuesday can be very different, so workers often talk about their *average* tips.

### 2. Pooled Tips and Tip-Outs

In many restaurants, all the tips go into one **tip pool**. At the end of the shift, the pool is shared. It can be shared **equally**, or **by hours worked**, so a person who worked longer gets more. Sharing by hours is sharing in a ratio, as in Unit 1.

Servers often also give a part of their tips to the kitchen staff, dishwashers or bussers. This is called a **tip-out**.

**Sharing a tip pool by hours**

Step 1. Add everyone's hours to get the total hours.
Step 2. Find the share for 1 hour: pool ÷ total hours.
Step 3. Multiply each person's hours by the share for 1 hour.
Step 4. Check: the shares add up to the pool.

**Example 4**

Four servers at a café in Cochrane share a tip pool of $612.00 from Saturday.

(a) How much does each server get if they share the pool equally?

(b) The café shares the pool by hours worked. Complete the table.

| **Server** | Mia | Jon | Priya | Sam | **Total** |
|---|---|---|---|---|---|
| **Hours** | 8 | 6 | 5 | 5 | |
| **Share** | | | | | |

**Example 5**

The bar at a Red Deer restaurant collects $825.00 in tips on a Friday night. First, 30% of the tips go to the kitchen staff. The rest is shared equally by the 3 bartenders. How much does each bartender get?

### 3. Bonuses

A **bonus** is extra money paid for good work, for reaching a goal, or for agreeing to stay with a company. A bonus can be:

- a **lump sum**: one set amount, like $500;
- a **percent** of your salary, wages or sales, like 4% of your yearly salary;
- an amount **per sale or per item**, like $15 for each phone plan you sell.

Year-end bonuses, signing bonuses (for joining or coming back) and referral bonuses (for bringing in a new worker) are common. A bonus is part of gross pay.

| **Bonuses** | percent bonus = amount × bonus rate | per-item bonus = number of items × bonus per item |
|---|---|---|

**Example 6**

(a) Kyle is a parts manager at a car dealership. He earns $46 800 a year. His year-end bonus is 4% of his salary. Find his bonus.

(b) A phone store pays a bonus for each plan sold: $8 for a Basic plan, $15 for a Plus plan, and $30 for a Premium plan. In March, Erin sold 12 Basic, 7 Plus and 4 Premium plans. Find her bonus.

**Example 7**

A hotel offers its housekeepers a choice of holiday bonus: a lump sum of $750, or 3% of their pay for the year. Rosa earned $28 600 this year. Which bonus should she choose, and how much more is it?

### 4. Shift Premiums, Danger Pay and Isolation Pay

Some hours or some jobs are harder, so employers pay extra for them.

- A **shift premium** is extra pay per hour for working evenings, nights or weekends. Nurses, health-care aides, security guards, factory and warehouse workers often get one. It is paid only for the hours in that time (for example, every hour after 6:00 p.m.).
- **Danger pay** (or hazard pay) is extra pay for risky work, such as fighting wildfires or working on power lines.
- **Isolation pay** is extra pay for working far from towns, such as at a northern mine or an oil sands camp. It can be per hour, per day or per month.

Extra pay can be a dollar amount per hour, or a percent more per hour.

**Extra pay per hour**

$ per hour: premium pay = premium hours × premium per hour (add it to the regular pay)

percent: new hourly rate = wage × (1 + rate) e.g. 30% more: wage × 1.30

**Watch out.** A shift premium is paid only on the hours inside the premium time, not on the whole shift. Count those hours carefully on the timesheet.

**Example 8**

Sana is a health-care aide at a seniors' home in Calgary. She earns $21.60/h, plus an evening premium of $2.50/h for every hour she works after 6:00 p.m. Her breaks are paid. Complete her timesheet and find her gross pay for the week.

| Day | Start | End | Hours worked | Hours after 6:00 p.m. |
|---|---|---|---|---|
| Monday | 2:00 p.m. | 10:00 p.m. | | |
| Tuesday | 10:00 a.m. | 6:00 p.m. | | |
| Wednesday | 3:00 p.m. | 11:00 p.m. | | |
| Friday | 4:30 p.m. | 11:30 p.m. | | |
| | | **Total** | | |

**Example 9**

(a) A wildfire crew member earns $24.00/h. For hours on the fire line, she gets danger pay of 30% more per hour. How much does she earn for an 8-hour day on the fire line? How much of that is danger pay?

(b) A cook at a northern mine camp earns $5850 a month. He can choose isolation pay of $650 a month, or 10% of his monthly pay. His contract is 4 months long. Which choice pays more, and by how much?

### 5. Putting It All Together: Total Gross Pay

| Extra | Jobs that often get it | Good for the worker | Not so good |
|---|---|---|---|
| Tips | servers, bartenders, stylists, drivers | good service can mean a lot more money | tips change from day to day; slow days pay less |
| Bonus | sales, office jobs, trades, hotels | a reward for working hard or staying | often not promised; can change each year |
| Shift premium | nurses, aides, guards, factory work | more pay per hour | late hours are hard on sleep and family time |
| Danger or isolation pay | firefighting, power lines, mines, camps | much higher pay | risky, or far from home for weeks |

To find total gross pay, find each kind of pay on its own, then add. A table keeps it organized.

Step 1: Regular pay = regular hours × wage. Overtime pay = overtime hours × 1.5 × wage (Lesson 3).
Step 2: Add each extra: premiums, tips, bonuses.
Step 3: Add everything to get the gross pay.

**Watch out.** Overtime is 1.5 times the *base* wage. A shift premium in dollars per hour is added on top, the same amount for every premium hour (overtime or not), unless the question says something else.

**Example 10**

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Diego is a cook at a hotel in Banff. He earns $19.00/h. This week he worked 46 hours, and never more than 8 hours in a day. He gets a night premium of $1.50/h, and 20 of his hours were night hours. His share of the kitchen tip-out was $142.75. Complete his pay summary.

| Type of pay | Work | Amount |
|---|---|---|
| Regular pay | | |
| Overtime pay | | |
| Night premium | | |
| Tip-out share | | $142.75 |
| **Gross pay** | | |

## Lesson 7: Deductions and Net Pay: Reading a Pay Stub

*Why your paycheque is smaller than what you earned, and how to check every line of your pay stub.*

### 1. Gross Pay, Deductions and Net Pay

Your **gross pay** is everything you earned in a pay period: wages, overtime, commission, bonuses. You never get all of it. Before you are paid, your employer takes some money off. Each amount taken off is a **deduction**.

The money that is left is your **net pay**, also called **take-home pay**. It is the amount that goes into your bank account. Each payday you also get a **pay stub** (or pay statement). It lists your gross pay, every deduction, and your net pay.

**Net pay**

total deductions = add up every deduction  net pay = gross pay − total deductions

**Watch out.** Once deductions come off, net pay is always less than gross pay. When you plan how to spend or save, use your *net* pay. That is the money you actually get.

**Example 1**

Kai works at a coffee shop in Cochrane. His gross pay for two weeks is $684.00. His deductions are CPP $40.70, EI $11.22, income tax $68.40, and $6.50 for his staff apron. Find his total deductions and his net pay.

### 2. Required Deductions: CPP, EI and Income Tax

By law, your employer must take three deductions off your pay and send them to the government (the **Canada Revenue Agency**, or **CRA**):

- **CPP** (**Canada Pension Plan**): money saved for when you retire. Your employer puts in the same amount as you.
- **EI** (**Employment Insurance**): if you lose your job (and it is not your fault), or you take time off for a new baby, EI pays you part of your wages for a while.
- **Income tax**: pays for things like hospitals, schools and roads. Part goes to the federal government and part to the province.

Each one is a percent of your pay. In this booklet, every question gives you the rates to use. They are **example rates**.

**Real life.** Real CPP, EI and income tax amounts come from CRA payroll tables or an online payroll calculator. They follow yearly rules. For example, there is no CPP on the first part of your yearly income, there is a most you can pay each year, and income tax depends on how much you earn in the whole year. Many students with part-time jobs get some or all of their income tax back when they file a tax return in the spring.

**A deduction that is a percent**

deduction = gross pay × rate (as a decimal)  e.g. CPP at 5.95%: gross pay × 0.0595

Round each deduction to the nearest cent *before* you add the deductions.

**Watch out.** To change a percent to a decimal, move the decimal point 2 places left: 5.95% = 0.0595 (not 0.595), 1.64% = 0.0164, 15% = 0.15.

**Estimate.** 1% of your pay: move the decimal point 2 places left. On $720, 1% is $7.20. CPP (about 6%) is about $6 \times \$7.20 \approx \$43$. EI (less than 2%) is less than $14.40. If your answer is far from the estimate, check your decimal.

**Example 2**

Hana works at a hotel front desk in Banff. Her gross pay for two weeks is $1240.00. Use the example rates in the table. Find each deduction, her total deductions, and her net pay.

| Deduction | Rate | Amount |
|---|---|---|
| CPP | 5.95% | |
| EI | 1.64% | |
| Income tax | 15% | |
| **Total deductions** | | |
| **Net pay** | | |

**Example 3**

Noah is 16. He works 18 hours one week at a garden centre in Calgary and earns Alberta's student wage of $13.00 an hour (as of 2026; rates can change). His deductions are CPP 5.95%, EI 1.64% and income tax 6%.

(a) Find his gross pay, each deduction, and his net pay.

(b) Noah wants an e-scooter that costs $650.00 with tax. If he saves all his net pay, how many weeks will it take?

### 3. Other Deductions and Percent of Gross Pay

Many jobs have **other deductions**. Some are a fixed dollar amount each payday and some are a percent:

- a **health and dental plan** (you and your employer share the cost),
- a **uniform** or safety gear, sometimes paid off over a few paycheques,
- a **charitable donation** you choose, such as to a food bank,
- parking, or a staff social club.

Every deduction goes into the total deductions.

To compare pay, it helps to know what *percent* of your gross pay you keep. This is the same "part ÷ whole × 100" you used in Unit 1.

**Percent of gross pay**

$$\text{take-home percent} = \frac{\text{net pay}}{\text{gross pay}} \times 100 \qquad \text{deductions percent} = \frac{\text{total deductions}}{\text{gross pay}} \times 100$$

**Finding a rate:** $\text{rate} = \dfrac{\text{deduction}}{\text{pay it was taken from}} \times 100$ The two percents add to 100%.

**Example 4**

Sofia drives a delivery van in Lethbridge. Her gross pay this week is \$960.00. Her deductions are CPP 5.95%, EI 1.64%, income tax 16%, a health plan of \$14.25, \$20.00 toward her safety boots, and a \$5.00 donation to the food bank. Find her net pay.

**Example 5**

(a) Use Example 4. What percent of her gross pay does Sofia take home? What percent goes to deductions? Round to the nearest tenth.

(b) Marcus's pay stub shows income tax of \$97.20 on gross pay of \$810.00. What income tax rate is he paying?

### 4. Before-Tax Deductions and Taxable Income

Some deductions come off *before* income tax is worked out. These are **before-tax deductions**: **union dues** (a fee for belonging to a union), a **company pension plan**, and **RRSP** savings (Registered Retirement Savings Plan) taken off at work. You pay income tax only on what is left. That amount is your **taxable income**.

**Taxable income**

taxable income = gross pay − before-tax deductions  income tax = taxable income × tax rate

CPP and EI are still worked out on the *gross* pay. If there are no before-tax deductions, taxable income = gross pay.

Step 1: Find each before-tax deduction.
Step 2: Taxable income = gross pay − before-tax deductions.
Step 3: CPP and EI on the gross pay; income tax on the taxable income.
Step 4: Add *all* the deductions. Step 5: Net pay = gross pay − total deductions.

**Watch out.** Before-tax deductions are still deductions. They come out of your pay, so count them in the total deductions.

**Example 6**

Ethan is a first-year carpentry apprentice in Edmonton. His gross pay for two weeks is \$2080.00. His before-tax deductions are union dues of 1.5% of gross pay and \$50.00 into his RRSP. His other deductions are CPP 5.95%, EI 1.64%, and income tax 18% of his taxable income.

(a) Find his taxable income and his net pay.

(b) How much less income tax does Ethan pay because of his before-tax deductions?

### 5. Reading and Checking a Pay Stub

A pay stub shows the **pay period** (the dates you worked), your hours and rate, your **earnings** (each kind of pay and the gross pay), each deduction, the total deductions, and the net pay. Many stubs also have a **year-to-date (YTD)** column: the total so far this year.

Mistakes happen on pay stubs, and it is your money, so check yours every payday:

1. Check the hours against your own record, and hours × rate for each earnings line.
2. Add the earnings to check the gross pay.
3. Estimate, then check each deduction.
4. Add the deductions, and check that net pay = gross pay − total deductions.

**Example 7**

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Lucas works at a warehouse in Balzac. Overtime is paid at time and a half. Complete his pay stub.

| **Prairie Link Logistics**, Balzac, AB | | | | | | |
|---|---|---|---|---|---|---|
| Employee: Lucas Moreau Pay period: Oct. 5–11, 2026 | | | | | | |
| **Earnings** | | | | **Deductions** | | |
| **Description** | **Hours** | **Rate** | **Amount** | **Description** | **Rate** | **Amount** |
| Regular | 40 | $19.20 | | CPP | 5.95% | |
| Overtime | 4.5 | | | EI | 1.64% | |
| | | | | Income tax | 14% | |
| | | | | Health plan | | $11.80 |
| | | **Gross pay** | | **Total deductions** | | |
| | | | | | **Net pay** | |

**Example 8**

Aisha's own record shows she worked 38 hours this week. Here is her pay stub. Find the two mistakes. Then find her correct net pay and how much she was underpaid.

| **Foothills Pharmacy**, Okotoks, AB | | | | | | |
|---|---|---|---|---|---|---|
| Employee: Aisha Rahman | | | Pay period: Nov. 2–8, 2026 | | | |
| **Earnings** | **Hours** | **Rate** | **Amount** | **Deductions** | **Rate** | **Amount** |
| Regular | 36 | \$17.75 | \$639.00 | CPP | 5.95% | \$38.02 |
| | | | | EI | 1.64% | \$104.80 |
| | | | | Income tax | 12% | \$76.68 |
| | | | | Uniform | | \$8.00 |
| | | **Gross pay** | \$639.00 | **Total deductions** | | \$227.50 |
| | | | | | **Net pay** | \$411.50 |

## Lesson 8: Working Backwards: How Many Hours Do I Need?

*Turn a pay formula around to find the hours, the wage or the sales you need, then check by putting your answer back in.*

### 1. Turning the Pay Formula Around

You know how to find gross pay: multiply the hourly wage (the **rate**) by the hours worked. But often you know the pay you *want*, and you need something else: "I want \$400. How many hours is that?"

This is **working backwards**. You **rearrange** the formula. That means you write it a new way, so the thing you want to find is by itself on one side of the equal sign.

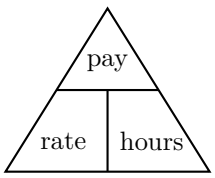
**The pay formula, three ways**

To find the pay: pay = rate × hours

To find the hours: hours = pay ÷ rate

To find the rate (wage): rate = pay ÷ hours

**Check:** put your answer back into pay = rate × hours. You should get the pay you started with.




**Formula triangle.** Cover the one you want. Side by side means ×; one above the other means ÷.

**Watch out.** Divide the *pay* by the *rate*, not the rate by the pay. Estimate first: \$520 at about \$16 an hour is about 30 hours. If your calculator says 0.03 hours, you divided the wrong way.

**Example 1**

(a) Marco earned \$446.25 for 25.5 hours of work at a garden centre. What is his hourly wage?

(b) Jonas earns \$16.25 an hour at a car wash. How many hours must he work to earn \$520.00?

**Round up.** When you need *at least* some amount of money, round the hours **up** to the next whole hour (or whole shift), even if the decimal is small: 41.1 h means you need 42 h. Rounding down would leave you a little short.

**Example 2**

Priya earns \$15.50 an hour at a grocery store. She wants to earn \$650.00 (before deductions) to buy a new phone.

(a) How many whole hours must she work?

(b) Her shifts are 6 hours long. How many shifts is that?

### 2. Working Backwards With Overtime

Remember the Alberta rule: hours over 8 in a day (or over 44 in a week, whichever gives more) are **overtime**, paid at 1.5 times the wage. If someone already works 8 hours every day, any extra hour added to a shift is an overtime hour.

| **How many overtime hours do I need?** | |
|---|---|
| Step 1: Find the pay for the regular hours. | Step 2: extra pay needed = target pay − regular pay |
| Step 3: overtime rate = 1.5× wage | Step 4: overtime hours = extra pay needed ÷ overtime rate |

**Example 3**

Kofi works 8 hours a day, Monday to Friday, at a warehouse. He earns \$20.00 an hour. Any extra hours he adds to a shift are overtime at time-and-a-half. How many overtime hours does he need to earn \$950.00 this week?

To find the *wage* when a pay stub has overtime in it, count the **paid hours**. Each overtime hour is paid like 1.5 regular hours (a double-time hour is paid like 2 hours).

| | |
|---|---|
| paid hours = regular hours + 1.5× overtime hours | wage = gross pay ÷ paid hours |

**Example 4**

Lena's pay stub shows 40 regular hours and 6 overtime hours (time-and-a-half). Her gross pay is $1029.00. What is her hourly wage?

### 3. How Much Do I Need to Sell?

A salesperson on commission earns a percent of what they sell. To reach a pay goal, they need to know how much to sell. Rearrange the commission formula the same way.

| | | |
|---|---|---|
| **Commission** | commission = rate × sales | sales needed = commission wanted ÷ rate |
| **Salary + commission:** first find the commission needed = target pay − salary. Then divide by the rate. | | |
| **Graduated commission:** use up the first level, then find the sales needed at the next rate. | | |

**Watch out.** Change the percent to a decimal: $8\% = 0.08$. $\$1200 \div 8 = \$150$ is wrong; $\$1200 \div 0.08 = \$15\,000$ is right. Sales needed are always much bigger than the commission.

**Example 5**

(a) Dana sells furniture and earns 8% commission. How much must she sell to earn $1200.00 in commission this month?

(b) Raj earns a salary of $1800.00 a month plus 4% commission on his sales of cell phone plans. How much must he sell to earn $3000.00 this month?

**Example 6**

Jasmine sells hot tubs. She earns 3% commission on the first $10 000 of her sales in a month and 5% on all sales over $10 000. How much must she sell to earn $1000.00 in commission?

### 4. Hours Needed After Deductions

Your pay cheque shows your **net pay**, not your gross pay. To buy something, you need enough *net* pay. So find how much you take home for each hour, then divide.

**Hours needed to take home an amount**

Step 1: total deduction rate = add the percents  Step 2: net per hour = wage $\times$ (1 $-$ total deduction rate)

Step 3: hours = net pay needed $\div$ net per hour, rounded **up**

Buying something in Alberta? net pay needed = price $\times$ 1.05 (5% GST)

**Note.** The CPP, EI and income tax rates in this booklet are simple examples. Real deductions come from CRA tables and change with your pay and the time of year.

**Calculator.** Don't round the net per hour too early. Write at least four decimals, or do it all in one line. In Example 8 below: 944.99 [÷] [(] 18.50 [×] 0.7741 [)] [ENTER].

**Example 7**

Ethan earns $16.00 an hour at a movie theatre. His deductions take about 20% of his gross pay. How many hours must he work to take home $600.00 for a trip to Jasper?

**Example 8**

Sofia earns $18.50 an hour. Her deductions are CPP 5.95%, EI 1.64% and income tax 15%. She wants a laptop that costs $899.99 in Calgary.

(a) What is the total cost of the laptop with GST?

(b) What percent of her gross pay does Sofia take home? What is her net pay for one hour?

(c) How many hours must she work to buy the laptop? Check your answer.

## Lesson 9: Unit 2 Review

*Every pay rule from the unit on one page, worked examples that mix the topics, and a practice unit test.*

## 1. The Unit on One Page

Use this page to study. Each row is one skill: what to do, and a short example.

| Skill | What to do | Example |
|---|---|---|
| **Time to decimal hours** | Minutes ÷ 60 gives part of an hour. Decimal part × 60 gives minutes. Benchmarks: 15 min = 0.25 h, 30 min = 0.5 h, 45 min = 0.75 h. | 7 h 40 min: $40 \div 60 = 0.67$, so 7.67 h 5.3 h: $0.3 \times 60 = 18$, so 5 h 18 min |
| **Hours on a shift** | Find the time from start to end (count to noon or to the next full hour). Subtract any **unpaid** break. 24-hour clock: subtract 12 from hours after 12 to get p.m. | 9:15 a.m. to 5:45 p.m. = 8 h 30 min; minus a 30 min unpaid lunch = 8 h |
| **Hourly wage** | gross pay = hours × rate. Alberta minimum wage: \$15.00/h (under 18 in school: \$13.00/h for the first 28 h a week). | 24.5 h at \$16.20/h: $24.5 \times 16.20 =$ \$396.90 |
| **Salary** | pay per period = yearly salary ÷ periods: weekly 52, biweekly 26, semi-monthly 24, monthly 12. Hourly equivalent = weekly pay ÷ hours a week. | \$45 760 a year, biweekly: $45\,760 \div 26 =$ \$1760.00 |
| **Overtime** | Time and a half = rate × 1.5; double time = rate × 2. **Alberta**: hours over 8 in a day, or over 44 in a week. Use whichever gives **more** overtime hours. | \$18.00/h: overtime rate $18 \times 1.5 =$ \$27.00/h |
| **Piecework and contracts** | Piecework: number of pieces × pay per piece. Contract: profit = price − costs (materials, helpers); effective hourly rate = profit ÷ hours. | \$2400 job, \$600 materials, 30 h: $1800 \div 30 =$ \$60.00/h |
| **Commission** | Straight: sales × rate. Salary + commission: add them. Graduated: a different rate on each part of the sales. Over a quota: commission only on sales above the quota. | 4% of \$12 500: $12\,500 \times 0.04 =$ \$500.00 |
| **Tips, bonuses, premiums** | Add to the gross pay. Pooled tips: total ÷ total hours = tips per hour, then × each person's hours. Shift premium: extra \$ per hour for those hours. | \$2.00/h premium on 10 night hours: +\$20.00 |
| **Deductions and net pay** | Each deduction = pay × rate, or a set amount (health plan). CPP and EI: on the gross. Income tax: on the **taxable income** = gross − before-tax deductions (union dues, pension, RRSP). net pay = gross − total deductions. | \$800 gross, \$184 deductions: net = \$616.00 ($616 \div 800 = 77\%$ take-home) |
| **Working backwards** | Rearrange: hours = pay ÷ rate; rate = pay ÷ hours; sales = commission ÷ rate. Round hours **up**. Check by putting the answer back in. | \$540 at \$18.00/h: $540 \div 18 = 30$ h. Check: $30 \times 18 = 540$ ✓ |

**Remember. Gross pay** is what you earn; **net pay** (take-home pay) is what is left after deductions. Change minutes to decimal hours *before* you multiply (6 h 45 min is 6.75 h, not 6.45 h). Round money to the cent only at the **end**. Deduction rates in this booklet are examples; real ones come from CRA tables and change every year.

## 2. Hours, Wages and Salaries

### Example 1

(a) Change each time.
(i) 2 h 40 min to decimal hours _____
(ii) 4.2 h to hours and minutes _____

(b) Leah works at a grocery store from 08:45 to 15:15 (24-hour clock) with a 30 min unpaid lunch. She earns $15.75/h. What is her gross pay for the shift?

### Example 2

Ines is offered a job as an office manager with a salary of $48 360 a year. She would work 40 h a week.

(a) How much is each pay if she is paid biweekly? semi-monthly?

(b) What is her hourly equivalent?

(c) Another office pays $23.75/h for 37.5 h a week. Which job pays more in a year, and by how much?

## 3. Overtime

**Watch out.** In Alberta, check the overtime **both** ways: add up the hours over 8 each day, and find the hours over 44 for the week. Use the **bigger** number. Regular hours = total hours − overtime hours.

### Example 3

Tyler works in a warehouse for $18.40/h. Overtime is paid at time and a half.

| **Day** | Mon | Tue | Wed | Thu | Fri | Sat | **Total** |
|---|---|---|---|---|---|---|---|
| Hours worked | 10 | 8 | 9 | 8 | 11 | 0 | |
| Hours over 8 | | | | | | | |

(a) Complete the table. How many overtime hours does Tyler get?

(b) Find his regular pay, his overtime pay and his gross pay.

(c) The next week Tyler works 9 h a day Monday to Friday and 6 h on Saturday. How many overtime hours does he get now? What is his gross pay?

### 4. Other Ways to Earn: Piecework, Contracts, Commission and Extras

**Example 4**

(a) Kiri plants trees in northern Alberta for \$0.16 a tree. One day she plants 1850 trees in 9.5 h. What does she earn, and what is that per hour?

(b) Owen bids \$1400 to stain a fence and deck. Stain and supplies cost him \$265. He pays a helper \$20.00/h for 12 h, and he works 16 h himself. What is his profit, and his effective hourly rate?

(c) Give one disadvantage of being paid by the job (a contract) instead of by the hour.

**Example 5**

Three salespeople are paid in different ways. Find each person's earnings for the month.

(a) Hana sells phones for a **straight commission** of 6%. Her sales are \$14 250.

(b) Diego earns \$1800 a month plus 2.5% on sales **over a quota** of \$20 000. His sales are \$34 600.

(c) Sofia earns a **graduated commission**: 3% on the first \$10 000 of sales and 5% on the rest. Her sales are \$26 400.

**Example 6**

(a) Three servers pool their tips and share them by hours worked. The tips total $630. Ana worked 18 h, Ben 12 h and Cole 15 h. How much does each get?

(b) Jas is a care aide who earns $24.80/h plus a $2.25/h night-shift premium. This week he works 36 h, all on nights, and gets a $150 bonus for perfect attendance. What is his gross pay?

## 5. Deductions and Net Pay

**Example 7**

Maya works 38 h at $19.50/h. Complete her pay stub. Use CPP 5.95%, EI 1.64% and income tax 15% of taxable income. She also pays $12.50 in union dues (a before-tax deduction).

| **Earnings** | | **Deductions** | |
|---|---|---|---|
| Hours | 38 | CPP (5.95%) | |
| Rate | $19.50 | EI (1.64%) | |
| **Gross pay** | | Income tax (15%) | |
| Taxable income | | Union dues (before tax) | $12.50 |
| | | **Total deductions** | |
| | | **Net pay** | |

(a) Complete the pay stub.

(b) What percent of her gross pay does Maya take home? Why are gross pay and net pay not the same?

### 6. Working Backwards

**Watch out.** Write the rearranged formula first, then put in the numbers. When you need enough hours to reach an amount, round **up**. If you need 86.2 h, then 86 h is not enough: round up to 87 h.

**Example 8**

(a) Liam was paid \$612.00 for 36 h. What is his hourly rate?

(b) Grace earns a straight 4% commission. What must she sell to earn \$2000 in a month?

(c) Zara wants a \$1100 phone, plus 5% GST. After deductions she takes home \$13.40 for each hour she works. How many hours must she work?